Exp. No: 15

# ANALYSE DIFFERENT TYPES OF SERVERS USING WEBALIZE TOOL.

## AIM:

To analyse different types of web servers using the Webalizer web log analysis tool and study statistics such as hits, visits, pages, URLs, browsers and data transferred.

## ALGORITHM:

1. Start the system and install the Apache web server.
2. Start the Apache service and verify its operation.
3. Locate the web server access log file.
4. Install the Webalizer log analysis tool.
5. Create an output directory for the reports.
6. Run Webalizer to analyse the access log.
7. Open the generated report in a web browser.
8. Analyze the statistics of different web servers.
9. Record the observations and terminate the program.

## ~~PROGRAM~~:

PROCEDURE - Fedora Linux.

Step 1: Install Apache

Open the terminal:

```
sudo dnf install httpd
```

Start Apache: sudo systemctl start httpd

Enable Apache at startup: sudo systemctl enable httpd

Check the service status: sudo systemctl status httpd

Open the browser and enter:

http://localhost

You should see the Apache test page.

Step 2: Locate the Apache access log.

```
sudo ls -l /var/log/httpd/
```

The access log is normally located at:

```
/var/log/httpd/access_log
```

To view its contents:

```
sudo cat /var/log/httpd/access_log
```

Step 3: Install Webalizer

```
sudo dnf install webalizer
```

Step 4: Create an output directory

```
sudo mkdir -p /var/www/html/webalizer
```

Step 5: Generate the Webalizer report

```
sudo webalizer -o /var/www/html/webalizer /var/log/httpd/
                                                access_log
```

Step 6: View the report

Open the browser and enter: http://localhost/webalizer/

Step 7: Analyze different types of servers.

The following table lists common web servers and their typical access-log locations.

| Web server | Typical access log |
|---|---|
| Apache HTTP Server | /var/log/httpd/access_log |
| Nginx | /var/log/nginx/access.log |
| Lighttpd | /var/log/lighttpd/access.log |

For example, to install and start Nginx on Fedora:

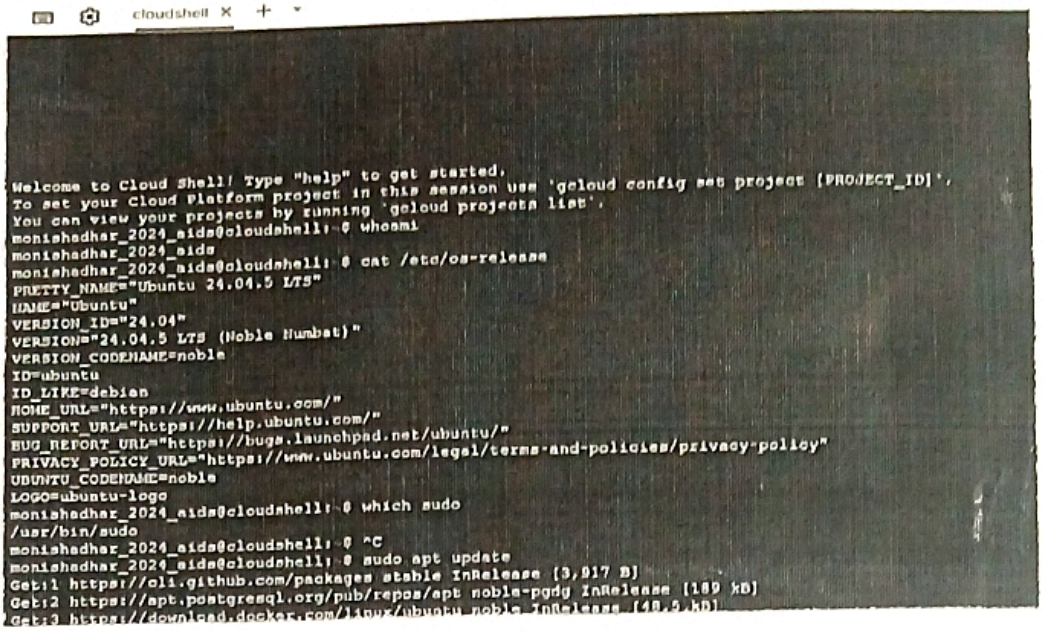
```
sudo dnf install nginx
sudo systemctl start nginx
sudo webalizer -o /var/www/html/webalizer-nginx
/var/log/nginx/access.log
```

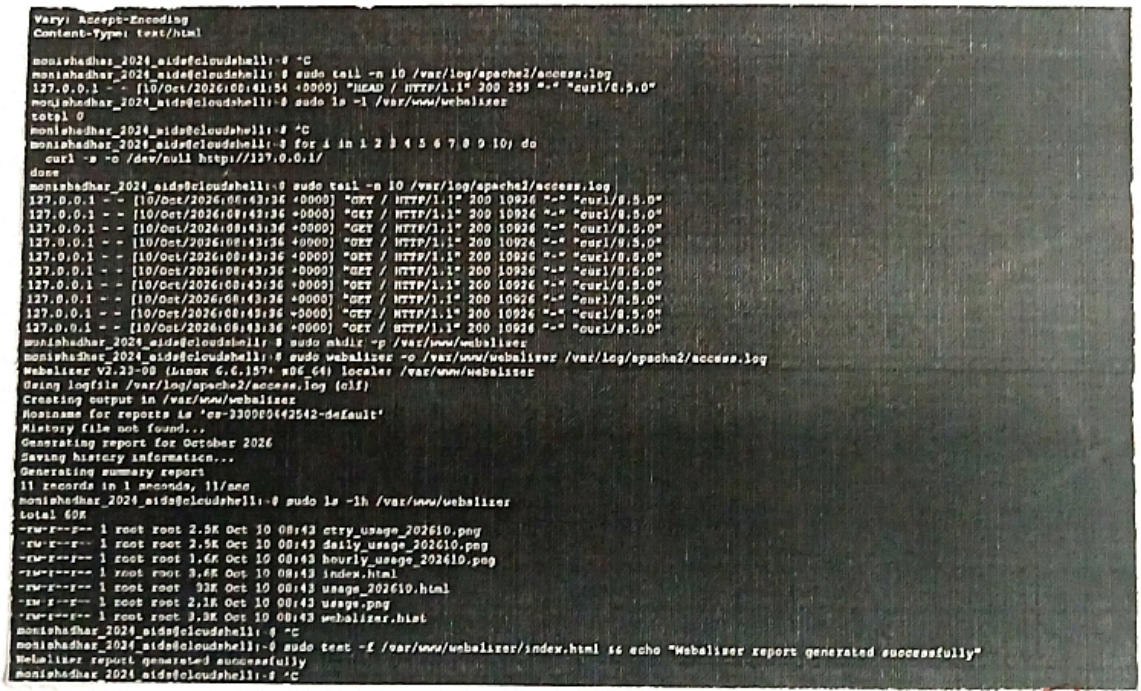
Then access:

```
http://localhost/webalizer-nginx/
```

RESULT:

Thus, the Webalizer tool was successfully used in Fedora to analyze web server log files and obtain statistics such as hits, visitors, pages, URLs and data transferred.


cloudshell
Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use 'gcloud config set project [PROJECT_ID]'.
You can view your projects by running 'gcloud projects list'.
monishadhar_2024_aids@cloudshell:~$ whoami
monishadhar_2024_aids
monishadhar_2024_aids@cloudshell:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.5 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
monishadhar_2024_aids@cloudshell:~$ which sudo
/usr/bin/sudo
monishadhar_2024_aids@cloudshell:~$ ^C
monishadhar_2024_aids@cloudshell:~$ sudo apt update
Get:1 https://cli.github.com/packages stable InRelease [3,917 B]
Get:2 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [189 kB]
Get:3 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]


Vary: Accept-Encoding
Content-Type: text/html
monishadhar_2024_aids@cloudshell:~$ ^C
monishadhar_2024_aids@cloudshell:~$ sudo tail -n 10 /var/log/apache2/access.log
127.0.0.1 - - [10/Oct/2026:08:41:54 +0000] "HEAD / HTTP/1.1" 200 259 "-" "curl/8.5.0"
monishadhar_2024_aids@cloudshell:~$ sudo ls -l /var/www/webalizer
total 0
monishadhar_2024_aids@cloudshell:~$ ^C
monishadhar_2024_aids@cloudshell:~$ for i in 1 2 3 4 5 6 7 8 9 10; do
curl -s -o /dev/null http://127.0.0.1/
done
monishadhar_2024_aids@cloudshell:~$ sudo tail -n 10 /var/log/apache2/access.log
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
127.0.0.1 - - [10/Oct/2026:08:43:36 +0000] "GET / HTTP/1.1" 200 10926 "-" "curl/8.5.0"
monishadhar_2024_aids@cloudshell:~$ sudo mkdir -p /var/www/webalizer
monishadhar_2024_aids@cloudshell:~$ sudo webalizer -o /var/www/webalizer /var/log/apache2/access.log
Webalizer V2.23-08 (Linux 6.6.157+ x86_64) locale: /var/www/webalizer
Using logfile /var/log/apache2/access.log (clf)
Creating output in /var/www/webalizer
Hostname for reports is 'cs-330095442542-default'
History file not found...
Generating report for October 2026
Saving history information...
Generating summary report
11 records in 1 seconds, 11/sec
monishadhar_2024_aids@cloudshell:~$ sudo ls -lh /var/www/webalizer
total 60K
-rw-r--r-- 1 root root 2.5K Oct 10 08:43 ctry_usage_202610.png
-rw-r--r-- 1 root root 2.5K Oct 10 08:43 daily_usage_202610.png
-rw-r--r-- 1 root root 1.6K Oct 10 08:43 hourly_usage_202610.png
-rw-r--r-- 1 root root 3.6K Oct 10 08:43 index.html
-rw-r--r-- 1 root root  93K Oct 10 08:43 usage_202610.html
-rw-r--r-- 1 root root 2.1K Oct 10 08:43 usage.png
-rw-r--r-- 1 root root 3.3K Oct 10 08:43 webalizer.hist
monishadhar_2024_aids@cloudshell:~$ ^C
monishadhar_2024_aids@cloudshell:~$ sudo test -f /var/www/webalizer/index.html && echo "Webalizer report generated successfully"
Webalizer report generated successfully
monishadhar_2024_aids@cloudshell:~$ ^C